1. What is Seaborn? How is it different from Matplotlib?

Seaborn is a Python data visualization library built on top of Matplotlib that provides a

high-level interface for drawing attractive and informative statistical graphics.

Differences:

Seaborn

High-level interface

Matplotlib

Built-in statistical plots

Low-level interface

Basic plotting functions

Better default themes & styles

Works well with Pandas DataFrames Works mainly with arrays/lists

Manual styling required

Seaborn simplifies complex visualizations and improves aesthetics compared to Matplotlib.

2. How to install Seaborn library?

Seaborn can be installed using:

pip install seaborn

Or using Anaconda:

conda install seaborn

To import:

import seaborn as sns

3. What is the difference between countplot() and barplot()?

countplot()

barplot()

Shows frequency/count of categorical

data

Shows aggregated value (mean by default)

Does not require y variable

Used for counting categories

Requires both x and y

Used for comparing numeric values across

categories

Example:

sns.countplot(x='gender', data=df)

sns.barplot(x='gender', y='salary', data=df)

4. How do you create a boxplot in Seaborn?

sns.boxplot(x='category', y='value', data=df)

A boxplot visualizes distribution using:

• Median

• Quartiles (Q1, Q3)

• Outliers

• Minimum & Maximum values

5. What is the use of hue parameter?

The hue parameter adds a third categorical variable to the plot using different colors.

Example:

sns.barplot(x='gender', y='salary', hue='department', data=df)

It helps compare sub-groups within categories.

6. What is kdeplot()?

kdeplot() (Kernel Density Estimate plot) visualizes the probability density function of a

continuous variable.

Example:

```
sns.kdeplot(df['age'])
```

It provides a smooth curve instead of histogram bars.

7. What is the use of jointplot()?

jointplot() shows the relationship between two variables along with their individual distributions.

Example:

```
sns.jointplot(x='age', y='salary', data=df, kind='scatter')
```

It displays:

- Scatter plot (center)
- Histogram or KDE (margins)

Used for correlation analysis.

8. What is pairplot()? What is its importance?

pairplot() creates pairwise scatter plots for all numerical columns in a dataset.

Example:

```
sns.pairplot(df)
```

Importance:

- Shows relationships between multiple variables
- Detects correlation patterns
- Identifies outliers
- Useful in exploratory data analysis (EDA)

9. What is the difference between distplot() and histplot()?

distplot()

Deprecated in newer versions Recommended replacement

histplot()

Combines histogram + KDE More flexible and customizable

Limited control

Supports multiple distributions

Example:

sns.histplot(df['age'], kde=True)

10. How do you create a heatmap?

sns.heatmap(data, annot=True, cmap='coolwarm')

A heatmap represents data in matrix form using colors.

11. What are the different categories available in Seaborn?

Seaborn plots are categorized into:

1. Relational plots

o scatterplot()

o lineplot()

2. Categorical plots

o barplot()

o countplot()

o boxplot()

o violinplot()

o stripplot()

o swarmplot()

3. Distribution plots

o histplot()

o kdeplot()

o ecdfplot()

4. Matrix plots

o heatmap()

o clustermap()

5. Regression plots

o regplot()

o lmplot()

6. Multi-plot grids

o pairplot()

o jointplot()

o FacetGrid()

12. How do we create a histplot()?

```
sns.histplot(df['age'], bins=20, kde=True)
```

Parameters:

• bins → number of intervals

• kde=True → adds density curve

13. How do you visualize the correlation matrix?

```
import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

corr = df.corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.show()
```

Steps:

1. Calculate correlation using df.corr()

2. Use sns.heatmap() to visualize

3. annot=True shows correlation values

This helps identify strong positive/negative relationships.

Sales Analysis Scenario

Problem:

You have monthly sales data for 5 years with order dates.

Q1: Which plot will you use to show trends?

To show trends over time, I would use a Line Plot (lineplot()).

Reason:

• Sales data is time-series data.

• A line plot effectively shows patterns, seasonality, upward/downward trends.

• It connects data points chronologically.

Implementation Example:

```
sns.lineplot(x='Order_Date', y='Sales', data=df)
```

If data is monthly aggregated:

```
df_monthly = df.groupby('Order_Date')['Sales'].sum().reset_index()

sns.lineplot(x='Order_Date', y='Sales', data=df_monthly)
```

Q2: How will you compare sales across different regions?

To compare sales across regions, I would use:

• Bar Plot (barplot()) → To compare average/total sales

• Line Plot with hue parameter → To compare trends region-wise

Example:

```
sns.barplot(x='Region', y='Sales', data=df)
```

Or:

```
sns.lineplot(x='Order_Date', y='Sales', hue='Region', data=df)
```

Reason:

• barplot() compares categorical values.

• hue allows region-wise differentiation using colors.

HR Dataset Scenario

Q1: Salary distribution by department – Which plot and why?

I would use:

• Boxplot

• Violin Plot

• Histplot (if overall distribution needed)

Preferred: Boxplot

```
sns.boxplot(x='Department', y='Salary', data=df)
```

Reason:

• Shows median, quartiles, and spread.

• Identifies salary variations across departments.

• Clearly highlights outliers.

Q2: How to check outliers in the dataset?

Outliers can be detected using:

1. Boxplot

2. IQR Method

3. Z-score Method

IQR formula:

IQR = Q3 - Q1

Lower Bound = Q1 - 1.5*IQR

Upper Bound = Q3 + 1.5*IQR

Q3: Importance of outliers & impact on ML models

Importance:

• May represent real rare cases (high performer salary)

• May indicate data entry errors

Impact on ML Models:

Model Type

Linear Regression Highly sensitive

Impact

KNN

Decision Trees

Distance affected

Less sensitive

Neural Networks Can distort learning

Outliers can:

• Skew mean and variance

• Reduce model accuracy

• Affect normalization and scaling

Proper handling improves model performance.

Healthcare Dataset – Cancer Analysis

Q1: How many people are affected with cancer?

Use:

```
df['Cancer'].value_counts()
```

To visualize:

```
sns.countplot(x='Cancer', data=df)
```

This shows number of affected vs non-affected individuals.

Q2: Gender-wise cancer analysis

```
sns.countplot(x='Gender', hue='Cancer', data=df)
```

This shows:

• Male affected count

• Female affected count

Alternatively:

```
pd.crosstab(df['Gender'], df['Cancer'])
```

Q3: Age group-wise cancer analysis (Child/Adult)

First categorize age:

```
df['Age_Group'] = df['Age'].apply(lambda x: 'Child' if x < 18 else 'Adult')
```

Then:

```
sns.countplot(x='Age_Group', hue='Cancer', data=df)
```

This helps identify vulnerable age groups.

Banking Dataset Scenario

Q1: How will you visualize correlation?

I will use a Heatmap.

```
corr = df.corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')
```

Q2: Which plot should be used and why?

Heatmap

Reason:

• Correlation matrix is numeric.

• Heatmap visually represents strength using color intensity.

• Easy to identify strong positive/negative relationships.

Q3: What is Correlation?

Correlation measures the strength and direction of linear relationship between two variables.

Range:

• +1 → Perfect positive correlation

• -1 → Perfect negative correlation

• 0 → No linear correlation

Q4: How to understand whether data is strongly correlated?

Correlation Value Interpretation

0.7 to 1 -0.7 to -1

Strong positive

Strong negative

0.3 to 0.7

0 to 0.3

Moderate

Weak

In heatmap:

• Dark red → Strong positive

• Dark blue → Strong negative

Also check:

• Multicollinearity using VIF

• Scatterplot patterns